



Course Syllabus

Foundation of Machine/DeepLearning with Python

COURSE INFORMATION

Course Number & Name:	CIST 007
Section Number:	CRN: 51138
Semester & Year:	Summer 2026
Lecture(days, time, location):	Monday through Thursday, 8AM – 12:15PM, In Person, CHE131
Lab (days, time, location):	Monday through Thursday, 8AM – 12:15PM, In Person, CHE131
Final Date and Time:	
Drop Policy:	
Prerequisites:	CIST 005A

INSTRUCTOR INFORMATION

Name:	Samira Pakravan, PhD
E-mail:	samira.pakravan@wvm.edu
Office hours (days, time, loc):	After our class or Online via Zoom by appointment

COURSE MATERIALS (NOT REQUIRED)

Title:	Machine Learning with PyTorch and Scikit-Learn: Develop machine learning and deep learning models with python
Author:	Sebastian Raschka
Publisher:	ISBN: 978-1801816380

***Recommended Materials and Resources**

- PyTorch Documentation: <https://pytorch.org/docs/>
- Scikit-Learn Documentation: <https://scikit-learn.org/stable/documentation.html>
- Python Documentation: <https://docs.python.org/3/>
- Jupyter Notebooks for interactive coding and experimentation



COURSE DESCRIPTION

This course introduces the foundations of machine learning and deep learning using Python. Students will learn core concepts underlying supervised and unsupervised learning, model training, evaluation, and optimization. The course emphasizes hands-on implementation using Scikit-Learn for classical machine learning methods and PyTorch for deep learning models.

Topics include data preprocessing, regression, classification, neural networks, model evaluation, and introductory deep learning architectures. The course is designed for beginners and focuses on developing both conceptual understanding and practical programming skills needed to apply machine learning techniques to real-world problems.

STUDENT LEARNING OUTCOMES

By successfully completing this course, students will be able to:

- Use Python to implement machine learning and deep learning workflows.
- Understand and apply fundamental machine learning concepts, including supervised and unsupervised learning.
- Preprocess and analyze data for use in machine learning models.
- Build, train, and evaluate machine learning models using Scikit-Learn.
- Implement basic neural networks using PyTorch.
- Interpret model performance using appropriate evaluation metrics.
- Apply machine learning and deep learning techniques to practical problem-solving tasks.
- Develop computational thinking skills relevant to data-driven and AI-based applications.

GRADING

A	90 - 100
B	80 - 89
C	70 - 79
D	60 - 69
F	< 60

- Quizzes ~20%
- Projects ~50%
- HW/Programs ~30%



Topics

Date	Topics
Week 1 (6/15-6/18)	Intro, Prep Software Environment NumPy, Matplotlib Pandas GitHub
Week 2 (6/22-6/25)	Intro to ML, Linear Regression (Cost Function, Gradient Descent) Multiple Linear Regression, Polynomial Regression Bias Variance Trade off & Cross Validation & Feature Scaling & Regularization Grid Search, Feature Engineering & Data Preparation
Week 3 (6/29-7/2)	Logistic Regression, Classification Metrics KNN - KNearest Neighbors, Support Vector Machine, Naive Bayes Classification Tree Based Methods, Random Forest , Boosting Methods
Week 4 (7/6-7/10)	Unsupervised Learning, K-Means Clustering Supervised learning Project
Week 5 (7/13-7/16)	Pytorch Feed Forward Neural Networks Convolutional Neural Networks Project Prep
Week 6 (7/20-7/23)	Convolutional Neural Networks RNN Project Presentation



CLASS ATTENDANCE

Students are expected to attend all sessions of each class. Instructors may drop students from the class if they fail to attend the first class meeting, or when accumulated unexcused hours of absences exceed ten percent of the total number of hours the class meets during the semester. Moreover, an instructor may drop from the class any student who fails to attend at least one class session during the first three weeks of instruction.

CANVAS INFORMATION

Canvas is a course management system adopted by the WVMCCD for all classes. When you log into the system, you will see a listing of classes that you are teaching (faculty) or a list of classes that you are taking. <https://wvm.instructure.com/login/canvas>

CHEATING POLICY

Dishonesty includes but is not limited to in-class cheating, out-of-class cheating, plagiarism, knowingly assisting another student in cheating or plagiarism, or knowingly furnishing false information to college staff, faculty, administrators or other officials. Following are definitions of in-class cheating, out-of-class cheating, plagiarism, and furnishing false information. These are not all-inclusive and the list itself is not meant to limit definition of cheating to just those mentioned.

- a. In-class cheating: during an examination or on any work for which the student will receive a grade or points, unauthorized looking at or procuring information from any unauthorized sources, or any other student's work.
- b. Out-of-class cheating: unauthorized acquisition, reading or knowledge of test questions prior to the testing date and time; changing any portion of a returned graded test or report and resubmitting as original work to be regraded; or presenting the work of another as one's own for a grade or points.
- c. Plagiarism: unauthorized use of expression of ideas from either published or unpublished work(s) as a student's own work for a grade in a class. This also includes the violation of copyright laws, including copying of software packages.
- d. Furnishing false information: forgery, falsification, alteration or misuse of college documents, records, or identification in class or in laboratory situations.

CODE OF STUDENT CONDUCT

The District shall enforce a student code of conduct the purpose of which is to promote and maintain orderly conduct of a responsible student body in a manner compatible with the District and College function as an educational institution (Education Code 76030)

http://www.westvalley.edu/documents/services/Policy/5.19_STUDENT_CONDUCT_POLICY.pdf



DISABILITY STATEMENT

ADA Statement: The American with Disabilities Act (ADA) is a federal anti-discrimination statute that provides comprehensive civil rights protection for persons with disabilities. Among other things, this legislation required that all students with disabilities be guaranteed a learning environment that provides for reasonable accommodation of their disabilities. Disability and Educational Support Program (DESP) supports students with learning and other disabilities to achieve their educational goals through comprehensive support and services including specialized counseling, learning disability services, alternate media and adapted physical education; call 408-741-2010; TTY: 408-741-2658.

SAFETY/EMERGENCY

It is the student's responsibility to know the evacuation procedures, evacuation route, and assembly area for this classroom. In case of an emergency, you are to follow the directions of your instructor. When directed to evacuate the classroom, *be sure to take all of your belongings when you leave* and remain with your class in the assembly area until you receive further directions.

WVM-Alert - Emergency Notification

Free WVM-Alert will text, email and call you to alert you to campus emergency situations. Sign in to www.wvm.edu/emergency and give us your contact information ASAP! If you don't sign up, you won't be notified!

FEES AND TUITION

Payment Plan Options:

West Valley College is committed to partnering with **Nelnet** to make your educational dreams possible. Starting Fall 2015 you will have the opportunity to select a payment plan to make your enrollment expenses more affordable. Additional information coming soon!

Note: Summer/Winter students are responsible for paying **all enrollment fees in full**.

Student Responsibility

Once a student has registered for classes, the student is considered officially enrolled.

It is the responsibility of the student to drop or withdraw completely otherwise the student will be accountable for all enrollment and related fees for the classes.

Failure to attend class meetings does not constitute withdrawal from the class and does NOT exempt a student from the obligation to pay all enrollment and related fees.

<http://westvalley.edu/admissions/fee-schedule.html>

Instructional Material Fee

It is the policy of the West Valley-Mission Community College District that the Governing Board may require students to provide instructional and other materials required for credit and non-credit



courses, provided that such materials are of continual value to a student outside of the classroom setting and provided that such materials are not solely or exclusively available from the District.

COUNSELING SERVICES

We encourage students to complete orientation and educational planning through the Counseling Services office in order to maximize their success. Data shows that students who have complete their educational planning and orientations before registering for classes have a higher success and completion rate than those who do not. For appointments or drop in information: call 408-741-2009 or just walk into the Counseling Building during open hours

STUDENT RIGHT TO KNOW

Please refer to the following links for information on the federally mandated requirements regarding a student's right to be informed about college data on graduation and completion rates and other information: <http://westvalley.edu/services/student-right-to-know/index.html>; <http://westvalley.edu/services/student-right-to-know/heoa/index.html>

TUTORING INFORMATION

Tutorial Services: Tutoring is available, without charge, to West Valley College Students. The service is provided by trained, qualified, students and professionals who have been recommended by faculty. Students are tutored in a drop-in, group, or individual environment depending on subject of interest. Most subjects are available. Students must be currently enrolled at West Valley College in the subject for which they request help. Study groups are welcomed and encouraged. To sign up for tutoring, come to the Tutorial Services office, or visit: <http://westvalley.edu/ss/tutorial/tutee.html>

Writing Center: The Writing Center is an open computer lab where students can get help with writing assignments for any class at WVC. Peer tutors and English faculty members are available to help you succeed. Open lab hours and one-on-one tutoring are available. The Writing Center provides: professional faculty members, drop-in individual tutoring, and a lab with computers, writing and reading resources, help with basic grammar, mechanics, essay organization, and much more. Visit the Writing Center, located in the Library, and please don't forget to bring your student ID# and a completed LS110E form. http://westvalley.edu/academics/language_arts/english/writingcenter.html

Math Resource Center (MRC)

The Math Resource Center (MRC), located in SM 5, supports West Valley College Students success in mathematic courses. Free tutoring is offered in all levels of mathematics. The MRC is staffed by math faculty and peer tutors. For more information please visit our website http://westvalley.edu/academics/science_math/mathematics/mrc.html



THE FAMILY EDUCATIONAL RIGHTS AND PRIVACY ACT (FERPA)

(20 U.S.C. § 1232g; 34 CFR Part 99) is a Federal law that protects the privacy of student education records. The law applies to all schools that receive funds under an applicable program of the U.S. Department of Education. For additional information, you may call 1-800-USA-LEARN (1-800-872-5327) (voice). Individuals who use TDD may call 1-800-437-0833.

SEXUAL HARASSMENT AND SEX DISCRIMINATION

In accordance with Title VII Section 1604, and Title IX of the 1972 Education Amendments, it is the policy of the West Valley-Mission Community College District to provide an educational, employment and business environment free of unwelcomed sexual advances, requests for sexual favors, and other verbal or physical conduct or communications constituting sexual harassment and/or sex discrimination as defined and otherwise prohibited by Federal and State law. Complaints by students or employees should be directed to Associate Vice Chancellor of Human Resources. The telephone number is (408) 741-2131.

Campus SAVE (Sexual Violence Elimination) Act: please refer to the following link for information on these federally mandated protections for victims of sexual violence.
<http://westvalley.edu/services/student-right-to-know/campus-save-act.html>

STUDENT HEALTH SERVICES

Clinical visits with a nurse; personal counseling for stress, anxiety, depression and relationship concerns; low cost prescriptions, immunizations; and support for victims of domestic violence, alcohol and substance abuse and sexual assault are all available through our excellent Student Health Services; call 408-741-2027; urgent contact line: 408-741-4000.

WEST VALLEY COLLEGE IS A NO-SMOKING AND DRUG-FREE CAMPUS

The West Valley-Mission Community College District policy 5.18.1 prohibits “the unlawful use, distribution, sale, or possession of alcohol, narcotics, dangerous or illegal drugs, or other controlled substances, as defined in California statutes, on District property or at any function sponsored by the District or colleges.” Also, the campus community voted it’s preference for a non-smoking environment, and the decision is reflected in District policy: The College’s non-smoking policy is enforced in all areas except for outer campus parking lots at least twenty five (25) feet away from buildings and pathways.



GRIEVANCE PROCESS

The grievance process is a formalized process to ensure the timely resolution of conflict at the lowest possible level. The first step is the informal resolution stage which involves the student who has a complaint and the staff member or specific group who is the other party in the grievance. The student must notify the staff person or representative of a group that she/he wishes to make an appointment for an informal meeting to review an action within ten (10) days of its occurrence. In the absence of the instructor or staff person and after a good faith effort to make contact, the grievant may directly contact the department chair. Additional information is available from the Vice President of Student Services.

POLICY FOR COURSE REPETITION

Title 5 code 55040: District Policy for Course Repetition. A student may repeat any course in which a substandard final grade (D F, NP, or W) was earned. A course may be repeated only once under this policy for a total of two attempts. A student wishing to repeat a course for a 3rd attempt will be required to submit a Student Petition Form.

NON-DISCRIMINATION STATEMENT

The District, and each individual who represents the District, shall provide access to its services, classes, and programs without regard to national origin, religion, age, gender, gender identity, gender expression, sex, race or ethnicity, color, medical condition, genetic information, ancestry, sexual orientation, marital status, physical or mental disability, pregnancy, or military and veteran status, or because he/she is perceived to have one or more of the foregoing characteristics, or based on association with a person or group with one or more of these actual or perceived characteristics.